LEAKLISTEN · CONCEPT OPTIONS

LeakListen concept options

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DOCUMENT CONTROL

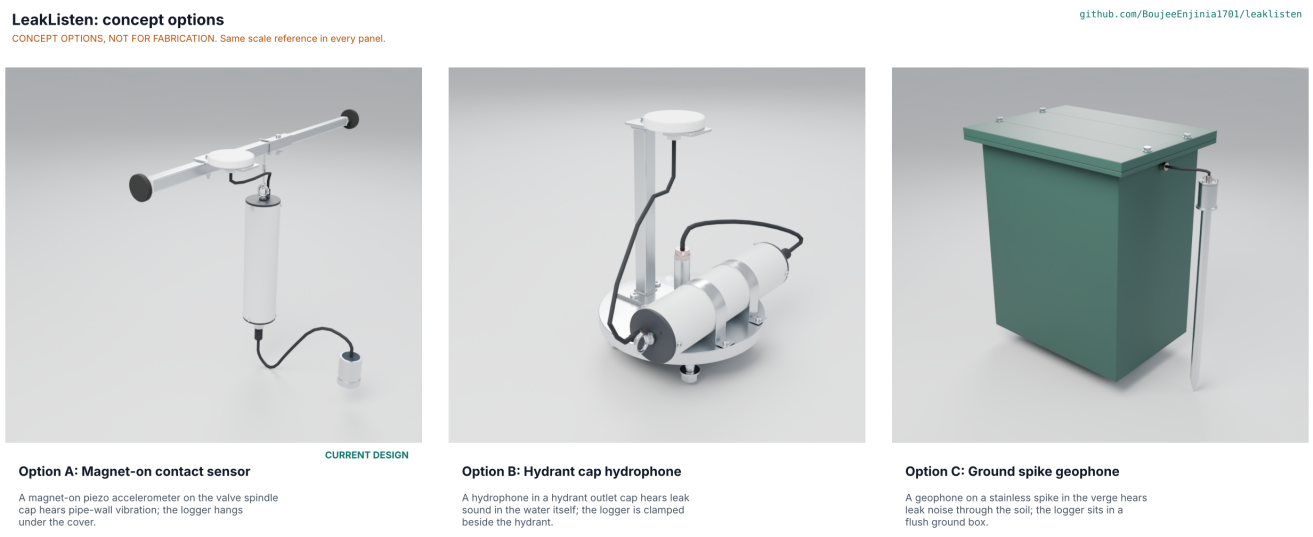
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
Source	docs/02a-concept-options.md @ a81e84a		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	Three options scored against LKL-URS-001 in the kit 1.9.1 upgrade; option A is the existing design, B and C added; every option modelled to D3 in cad/src/model.py

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Result: option A, the magnet-on contact sensor (the existing design), scored 155, option B, the hydrant cap hydrophone, 127, and option C, the ground spike geophone, 123, out of 205. Option A leads option B by 18 %, outside the 10 % band, so one option is recommended: **option A, with option B kept as the separate variant for plastic mains**. This is **Proposed, awaiting Amish** (LKL-DDR-004); nothing has been chosen. All three options are modelled to detail level D3 in `cad/src/model.py` ("Detail every option", Amish, 2026-10-24). The picture below is redrawn from those models by the coordinator's Blender run (`media/options.png` is not in the repo yet; see `docs/REVIEW.md`). Method: STANDARDS section 21.



The options

Option A: magnet-on contact sensor

A sealed piezo accelerometer puck on a 42 mm pot magnet snaps onto the valve spindle cap and listens to the pipe wall's vibration through the valve; a 63 mm PVC logger hangs under the cover from a neck bar and sends a 24-byte summary a night by LoRaWAN. This is the existing design (LKL-PRC-001) and does not change. It is placed from the surface, touches no water and weighs 0.96 kg (logger, cable and sensor); it hears about 185 m of iron main but only a few metres of plastic main, and its radio is at risk under an iron cover. Needs from a partner: valve and cover types, cover loss measurement.

Option B: hydrant cap hydrophone

A stainless blind cap bolts to a hydrant outlet and carries a piezoceramic tube hydrophone in the water column; the same logger lies along the cap in two clips, and a short mast carries the antenna to just under the hydrant cover. A hydrophone hears the leak noise as pressure in the water on any pipe material (commercial hydrophone loggers on hydrant caps are sold, see LKL-MKT-001), which is the answer to plastic mains (UR-002). It sits in drinking water (product approval, UR-012), needs a hydrant with an open valve to fill the barrel, weighs 6.0 kg in the model and its cap interface is an assumption: the model has a four-bolt flange, real outlets are mostly threaded (STM SB4). Needs from a partner: hydrant type, permission to open hydrants, a water approval route.

Option C: ground spike geophone

A stainless spike driven 0.5 m into the verge carries a geophone that hears the ground-borne noise of a leak, with its cable running to the same logger standing in a flush HDPE ground box with the antenna under a plastic lid. It needs no access to the pipe or the chamber and the antenna is not under iron, but the leak is heard only a few metres from the spike (STM SC1), the box is a 15.3 kg dig, and ground noise (traffic, trees) is likely to be high. Needs from a partner: utility records of where services are, permission to dig.

First-order numbers

Table 1. Options compared (D3 model figures from `cad/fidelity.json`).

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	142 (priced BOM, `bom/bom.csv`)	At least 106 from the shared priced lines (logger 97 and sensing proxy 9) plus the cap, gasket, bolts, mast and clips, unpriced (to assess)	At least 106 from the shared priced lines plus the box, lid, spike, plate and fasteners, unpriced (to assess)
Mass of the D3 model (kg)	1.3 (0.96 for logger, cable and sensor)	6.0	15.3
Reach for the reference leak	About 185 m on iron, 4 m on PVC	54 m on PVC at an assumed 0.1 Pa/rtHz background (none at 1 Pa/rtHz; 17 m at the lowest efficiency)	A few metres of main: +14 dB signal to noise at 3 m, -6 dB at 10 m
Energy or effort	1.25 mAh a day, 10 years or more	Same logger	Same logger
Parts in the D3 model (with fasteners and welds)	72 (24 fasteners)	89 (40 fasteners, 8 welds)	80 (38 fasteners)
STEM check	Open: 1 detail fixed (S3), no fails	Open: plausible rows SB1, SB3, SB4	Open: plausible rows SC1, SC2, SC3

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5). One criterion per user requirement in LKL-URS-001; cost, buildability, safety and maintenance are covered by UR-008, UR-015, UR-011 and UR-015.

Criterion	Source	Weight	A	B	C	Reason
Hears leaks before they surface	UR-001	3	3	4	1	A hears about 185 m on iron (at risk); B hears the water column directly; C hears only a few metres from the main (STM SC1)
Works on plastic mains too	UR-002	2	1	4	2	A hears 4 m on PVC; B 54 m on PVC at an assumed background; C couples through soil, any pipe, a few metres
Placed from the surface	UR-003	3	5	4	2	A lowers a puck by pole; B bolts a cap on a surface hydrant outlet; C digs a 0.42 x 0.30 x 0.52 m hole near buried services
Quick to place	UR-004	2	4	3	1	A 11 min estimate; B four M12 bolts and a 6 kg cap (not estimated); C a hole, a box and a spike
Reports by itself	UR-005	3	3	3	5	A and B transmit from under iron covers (range at risk); C's antenna sits under a plastic lid in the verge
Survives the chamber	UR-006	3	4	3	3	A is IP68 and floats; B sits on a cap under hydrant pressure and frost; C is buried in wet soil
Runs for years unattended	UR-007	3	4	4	4	All three use the same logger electronics and cell (10 years on energy)
Affordable for a small utility	UR-008	3	3	2	2	A USD 142 priced; B and C add a stainless cap or an HDPE box and a spike, unpriced (to assess), shared lines alone are USD 106
Raw sound stays on the device	UR-009	3	5	4	4	Same rule in all three; a hydrophone or geophone is a better microphone than a contact sensor
Open data and standard radio	UR-010	2	4	4	4	Same radio and payload
Safe to handle	UR-011	3	3	2	3	B: a cap on a pressurised hydrant is a hazard if opened; C: digging and a driven spike; A: magnet, cover lifting, lithium cell
Does not touch the water	UR-012	3	5	1	5	B's hydrophone sits in drinking water and needs product approval
Few false alarms	UR-013	2	3	3	2	C hears traffic and trees; B hears hydraulic transients; A is the quietest coupling but still hears pumps
Small and light	UR-014	2	5	3	1	Model mass 1.3 / 6.0 / 15.3 kg
Open and repairable	UR-015	2	4	3	4	B needs a machined 316 cap and potted hydrophone; A and C use stock sections and turned or moulded parts
Legal to place and to transmit	UR-016	2	4	3	4	B needs drinking-water product approval; C needs consent to dig; A needs permission to place only
Weighted total			**155**	**127**	**123**	

Weighted total = sum of score x weight over the 16 criteria (maximum 205). The scores are the author's judgement from the first-order numbers; they are not field data and the weights are the priorities in LKL-URS-001.

Recommendation

Option A, because it is the only option that is light, cheap, placed from the surface without touching the water, and already worked through to a priced, buildable design (155 against 127 and 123). Its weaknesses are the two the TRL 3 calculations already found: a short reach on plastic mains and a radio link that closes only with a gateway within about 0.5 km under an iron cover. **Option B's best idea**, a hydrophone on a hydrant cap, is the answer to the first weakness and is kept as the plastic-main variant (PR-002) rather than grafted onto option A, since a hydrophone changes the sensing principle and the water-contact rules. Option C is the weakest: the leak is heard only a few metres away. If Amish prefers a single product for all pipe materials, option B has the stronger acoustics (SB1) but sits in the water, weighs six times more and has an unpriced cap.

Recorded as "Proposed, awaiting Amish" in the decisions register (LKL-DEC-001) and argued in LKL-DDR-004.